NEAR-FIELD UAV LOCALIZATION VIA FAR-FIELD INITIALIZED BLOCK COORDINATE DESCENT

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Abstract

Accurate localization of unmanned aerial vehicles (UAVs) in the near-field regime is a critical yet challenging task. The problem is compounded by the spatially-variant Doppler effects, which arise from spherical wavefronts and the high mobility of UAV targets. Existing methods, predominantly designed for static scenarios, fail to jointly address these dynamic phenomena and are thus rendered ineffective. To address this, we propose a two-stage localization framework. To be specific, the first stage efficiently obtains a coarse estimate via a far-field approximation, which is then fed into a second stage for high-accuracy refinement. This refinement stage, based on a block coordinate descent (BCD) framework, uniquely operates on the exact near-field high-mobility signal model, enabling it to resolve the full set of coupled location and motion parameters. Simulation results demonstrate that the proposed method outperforms existing approaches in terms of estimation accuracy.

1 Introduction

Integrated sensing and communication (ISAC) is a key technology for future mobile networks, where the communication infrastructure can be exploited in various sensing tasks, such as object detection, environment reconstruction, and target localization [1–5]. Recently, with the rapid proliferation of unmanned aerial vehicles (UAVs) in low-altitude airspace, the localization of UAVs has emerged as a prominent application [6, 7]. Compared to traditional localization systems based on dedicated radar equipments and radio frequency (RF) receivers [8–10], the ISAC paradigm provides a cost-effective and wide-area solution for UAV localization, owing to the ubiquitous coverage and advanced processing capabilities of existing base stations (BSs).

In pursuit of higher data rates, the BSs are evolving towards large-scale arrays and high-frequency bands [11]. This trend has made near-field effects a non-negligible factor in the associated ISAC systems [12, 13]. As such, the conventional far-field assumption of planar wavefronts is invalidated, necessitating the adoption of spherical wavefront model [14]. However, this transition poses two challenges to the localization of high-mobility UAVs. Firstly, it induces a strong non-linear coupling between the range and angle of UAVs, rendering these location parameters inseparable within the phase information across the array [15]. Secondly, the motion of the UAVs results in a spatially-variant Doppler shift across the array, which is jointly determined by both radial and transverse velocities [16, 17]. Collectively, these challenges result in a deep interdependency among all motion and location parameters, which complicates an efficient and accurate parameter estimation.

To address the range-angle coupling inherent in near-field localization, a variety of approaches have been proposed. For example, in sparse recovery approaches [18, 19], distance-parameterized dictionaries are designed for iterative parameter estimation. Concurrently, subspace-based techniques utilize operators like oblique projection to separate signal subspaces and extract target locations [20–22]. However, since these approaches are predominantly designed for static scenarios, their estimation performance can be limited in high-mobility situations where complex Doppler effects disrupt the underlying signal model. While a few methods consider mobile target localization [23, 24], they typically trade accuracy for tractability by resorting to simplified models, which neglect the spatial variation of Doppler shifts or the impact of transverse velocity. Such simplifications introduce a model mismatch, imposing a performance ceiling on the achievable localization accuracy.

In this paper, we propose a near-field UAV localization method, which decomposes the highly-coupled parameter estimation problem into two stages. Specifically, the first stage deliberately employs a far-field approximation to obtain a coarse yet computationally efficient initial estimate. Building upon this, the second stage refines the initial estimated by systematically resolving the parameter interdependencies with a block coordinate descent (BCD) framework. The simulation results validate the effectiveness proposed method in terms of estimation accuracy.

2 System Model

As shown in Fig. 1, we consider a near-field UAV localization scenario, where a base station (BS) equipped with a N-element uniform linear array (ULA) aims to locate Q UAVs. The ULA is centered at the origin, and the position of the n-th array element is denoted by $\boldsymbol{p}_n = [d_n, 0]^{\mathrm{T}}$ with

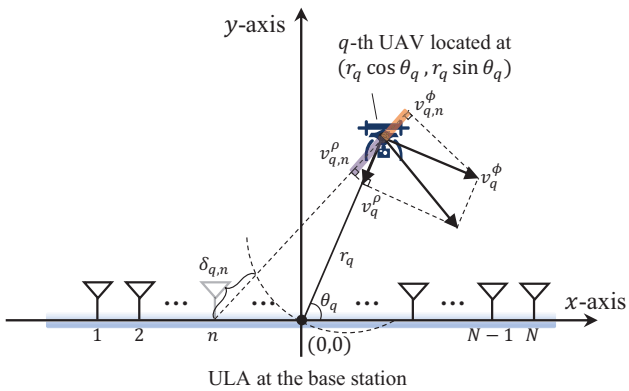

Fig. 1. Illustration of near-field UAV localization.

$d_n = (n - \frac{N+1}{2})\frac{\lambda}{2}$, $n = 1, 2, \cdots, N$. Here, $\lambda = \frac{f_c}{c}$ denotes the signal wavelength, f_c is the carrier frequency, c denotes the light speed, and $(\cdot)^{\mathrm{T}}$ represents the transpose. The state of the q-th UAV is characterized by its current location and motion direction. The location $\boldsymbol{u}_q = [r_q \cos\theta_q, r_q \sin\theta_q]^{\mathrm{T}}$ is parameterized by the range r_q and angle θ_q relative to the array center. Meanwhile, the motion of the q-th UAV is defined by its radial velocity v_q^{ρ} and transverse velocity v_q^{ϕ}. This motion results in an equivalent radial velocity relative to the n-th array element by projecting of the velocity vector onto the corresponding line-of-sight, which can be calculated as

$$\nu_{q,n} = \frac{r_q - \frac{n\lambda}{2}\cos\theta_q}{\|\boldsymbol{p}_n - \boldsymbol{u}_q\|} v_q^{\rho} + \frac{n\frac{\lambda}{2}\sin\theta_q}{\|\boldsymbol{p}_n - \boldsymbol{u}_q\|} v_q^{\phi}. \tag{1}$$

Here,

$$\|\boldsymbol{p}_n - \boldsymbol{u}_q\| = \sqrt{r_q^2 + d_n^2 - 2d_n r_q \cos\theta_q} \tag{2}$$

is the distance between the q-th UAV and the n-th array element of the BS.

To actively localize the UAVs, The BS transmits an orthogonal frequency division multiplexing (OFDM) waveform, comprising M symbols over K subcarriers. The symbol vector transmitted on the k-th subcarrier within the m-th OFDM symbol duration is denoted as $\boldsymbol{s}_{k,m} = [s_{k,m,1}, s_{k,m,2}, \cdots, s_{k,m,N}]^{\mathrm{T}}$, where each symbol $s_{k,m,n}$ is drawn from quadrature phase-shift keying (QPSK) constellation. After reflection from the UAVs, the received signal vector $\boldsymbol{y}_{k,m} \in \mathbb{C}^N$ at the BS is represented as

$$\boldsymbol{y}_{k,m} = \sum_{q=1}^{Q} \alpha_q e^{-\jmath 2\pi k \Delta \frac{2r_q}{c}} \left(\boldsymbol{a}_q \boldsymbol{a}_q^{\mathrm{T}} \odot \boldsymbol{d}_{q,m} \boldsymbol{d}_{q,m}^{\mathrm{T}}\right) \boldsymbol{s}_{k,m} + \boldsymbol{w}_{k,m}. \tag{3}$$

Here, α_q denotes the complex gain of the propagation path corresponding to the q-th UAV,

$$\boldsymbol{a}_q = \left[e^{-\jmath\frac{2\pi}{\lambda}\delta_{q,1}}, e^{-\jmath\frac{2\pi}{\lambda}\delta_{q,2}}, \cdots, e^{-\jmath\frac{2\pi}{\lambda}\delta_{q,N}}\right]^{\mathrm{T}} \tag{4}$$

is the spatial steering vector with

$$\delta_{q,n} = \sqrt{r_q^2 + d_n^2 - 2d_n r_q \cos\theta_q} - r_q \tag{5}$$

being the differential path length,

$$\boldsymbol{d}_{q,m} = [e^{\jmath\frac{2\pi}{\lambda}\nu_{q,1} m T_{\mathrm{s}}}, e^{\jmath\frac{2\pi}{\lambda}\nu_{q,2} m T_{\mathrm{s}}}, \cdots, e^{\jmath\frac{2\pi}{\lambda}\nu_{q,N} m T_{\mathrm{s}}}]^{\mathrm{T}} \tag{6}$$

denotes the Doppler steering vector, T_{s} is the OFDM symbol duration, $\odot$ is the Hadamard product, and $\boldsymbol{w}_{k,m} \sim \mathcal{CN}(0, \sigma^2\mathbf{I})$ denotes the additive white Gaussian noise vector.

Note that, the received signal model in (3) exhibits parameter coupling in the spatial and temporal domains owing to the spatial term $\boldsymbol{a}_q$ in (4) and Doppler term $\boldsymbol{d}_{q,m}$ in (6), respectively. This makes a direct, high-dimensional grid search computationally intractable.

3 Near-Field UAV Localization

In this section, we propose a two-stage hierarchical method that balances estimation accuracy and computational load. It consists of an initial coarse estimation stage, followed by a refinement stage employing block coordinate descent (BCD) framework.

3.1 Stage I: Initialization via Far-Field Approximation

The initial stage aims to efficiently obtain a coarse parameter estimate of $\{r_q, \theta_q, v_q^{\rho}\}$ by strategically simplifying the signal model (3) through a far-field approximation. While this introduces a deliberate model mismatch, it decouples parameters across the spatial, temporal, and frequency domains, enabling low-complexity one-dimensional parameter estimation.

To achieve this decoupling, the element-dependent phase variations are linearized in the spatial and Doppler domains, respectively. First, for the spatial domain, the differential path length $\delta_{q,n}$ in (5) is approximated by performing first-order Taylor expansion around the reference distance r_q as

$$\sqrt{r_q^2 + d_n^2 - 2d_n r_q \cos\theta_q} \approx r_q \sqrt{1 + \frac{d_n^2}{r_q^2} - \frac{2d_n}{r_q}\cos\theta_q} \tag{7}$$

and using the binomial approximation $\sqrt{1+x} \approx 1 + \frac{x}{2}$ for small value of x. Then, we have

$$\delta_{q,n} \approx -\left(n - \frac{N+1}{2}\right)\frac{\lambda}{2}\cos\theta_q, \tag{8}$$

which decouples the range and angle parameters within the spatial domain. As such, the spatial steering vector $\boldsymbol{a}_q$ in (4) reduces to its far-field counterpart

$$\dot{\boldsymbol{a}}_q = e^{-\jmath\pi\frac{N+1}{2}\cos\theta_q}\left[e^{-\jmath\pi\cos\theta_q}, e^{-\jmath\pi 2\cos\theta_q}, \cdots, e^{-\jmath\pi N\cos\theta_q}\right]^{\mathrm{T}}. \tag{9}$$

Second, for the Doppler domain, the antenna-specific radial velocity $\nu_{q,n}$ is approximated by the common radial velocity v_q^{ρ}. This approximation holds under the far-field assumption ($d_n \ll r_q$), where the distance $\|\boldsymbol{p}_n - \boldsymbol{u}_q\|$ is approximated by the range r_q. Substituting this into (1), the coefficients for v_q^{ρ} and v_q^{ϕ} approach 1 and 0, respectively, resulting in $\nu_{q,n} \approx v_q^{\rho}$.

Incorporating the approximations for $\delta_{q,n}$ and $\nu_{q,n}$ into (3), the received signal model can be approximated by

$$\boldsymbol{y}_{k,m} \approx \sum_{q=1}^{Q} \alpha_q e^{-\jmath 2\pi k \Delta \frac{2d_q}{c}} e^{\jmath \frac{2\pi}{\lambda} 2\nu_q^{\rho} m T_{\mathrm{s}}} \dot{\boldsymbol{a}}_q \dot{\boldsymbol{a}}_q^{\mathrm{T}} \boldsymbol{s}_{k,m} + \boldsymbol{w}_{k,m}. \quad (10)$$

The parameters for range r_q, angle θ_q, and radial velocity v_q^{ρ} are decoupled into distinct terms in (10), enabling an efficient, sequential estimation strategy. In particular, the angles can be first estimated via subspace-based method in the spatial domain as

$$\{\hat{\theta}_q\}_{q=1}^{Q} = \arg \underset{\theta}{\text{peak}} \left\{ \frac{1}{\|\boldsymbol{U}^{\mathrm{H}} \dot{\boldsymbol{a}}(\theta)\|_2^2} \right\}, \quad (11)$$

where $\boldsymbol{U}$ denotes the noise subspace of the sample covariance matrix $\hat{\boldsymbol{R}} = \frac{1}{KM} \sum_{k=1}^{K} \sum_{m=1}^{M} \boldsymbol{y}_{k,m} \boldsymbol{y}_{k,m}^{\mathrm{H}}$, and $\|\cdot\|_2$ denotes the Euclidean norm. Subsequently, for each estimated angle $\hat{\theta}_q$, the corresponding range $\hat{r}_q$ and radial velocity $\hat{v}_q^{\rho}$ are jointly estimated by locating the peaks of a 2D periodogram, formed by applying a 2D-fast Fourier transform (FFT) to the spatially filtered signal $z_{k,m}(\hat{\theta}_q) = \dot{\boldsymbol{a}}(\hat{\theta}_q)^{\mathrm{H}} \boldsymbol{y}_{k,m}$ as

$$(\hat{r}_q, \hat{v}_q^{\rho}) = \arg \max_{r, v^{\rho}} \left| \sum_{k=1}^{K} \sum_{m=1}^{M} z_{k,m}(\hat{\theta}_q) e^{-\jmath 2\pi (k \frac{2r}{c/\Delta f} + m \frac{2v^{\rho}}{\lambda/T_{\mathrm{s}}})} \right|^2. \quad (12)$$

This sequential process naturally provides paired parameter estimates $\{\hat{\theta}_q, \hat{r}_q, \hat{v}_q^{\rho}\}$ for the refinement stage.

3.2 Stage II: Refinement via Block Coordinate Descent

The coarse estimates obtained in Stage I are computationally efficient, but its accuracy is limited by the far-field approximation. In this regard, a high-accuracy refinement is performed in Stage II by directly optimizing the parameters using the exact near-field signal model in (3).

The refinement is formulated as a non-linear least squares (NLLS) problem, with the objective of minimizing the squared error between the received signal $\boldsymbol{y}_{k,m}$ and the reconstructed signal as

$$J(\{\boldsymbol{\xi}_q\}_{q=1}^{Q}) = \sum_{k=1}^{K} \sum_{m=1}^{M} \left\| \boldsymbol{y}_{k,m} - \sum_{q=1}^{Q} \tilde{\boldsymbol{y}}_{k,m}(\boldsymbol{\xi}_q) \right\|_{\mathrm{F}}^2, \quad (13)$$

where

$$\tilde{\boldsymbol{y}}_{k,m}(\boldsymbol{\xi}_q) = \alpha_q e^{-\jmath 2\pi k \Delta \frac{2r_q}{c}} \left(\boldsymbol{a}_q \boldsymbol{a}_q^{\mathrm{T}} \odot \boldsymbol{d}_{q,m} \boldsymbol{d}_{q,m}^{\mathrm{T}} \right) \boldsymbol{s}_{k,m} \quad (14)$$

represents the reconstructed signal from the q-th UAV, $\boldsymbol{\xi}_q = \left[\alpha_q, r_q, \theta_q, v_q^{\rho}, v_q^{\phi}\right]^{\mathrm{T}}$ denotes the corresponding parameter vector, and $\|\cdot\|_{\mathrm{F}}$ is the Frobenius norm. Instead of optimizing all variables in (13) simultaneously, which is computationally demanding, the BCD framework is employed to refine the parameters for each UAV target cyclically.

In particular, for the q-th UAV, the optimization subproblem is to minimize $J_q = \sum_{k,m} \|\boldsymbol{r}_{q,k,m} - \tilde{\boldsymbol{y}}_{k,m}(\boldsymbol{\xi}_q)\|_{\mathrm{F}}^2$, where the residual signal $\boldsymbol{r}_{q,k,m}$ is defined as

$$\boldsymbol{r}_{q,k,m} = \boldsymbol{y}_{k,m} - \sum_{q' \neq q} \tilde{\boldsymbol{y}}_{k,m}(\boldsymbol{\xi}_{q'}). \quad (15)$$

This residual effectively isolates the contribution of the q-th UAV, treating the interference from other UAVs as cancelled. The parameter blocks in $\boldsymbol{\xi}_q$ are then updated as follows.

1) Updating the Complex Gain

When other parameters are fixed, the signal model is linear with respect to α_q, i.e., $\tilde{\boldsymbol{y}}_{k,m}(\boldsymbol{\xi}_q) = \alpha_q \boldsymbol{z}_{q,k,m}$, where

$$\boldsymbol{z}_{q,k,m} = e^{-\jmath 2\pi k \Delta \frac{2r_q}{c}} \left(\boldsymbol{a}_q \boldsymbol{a}_q^{\mathrm{T}} \odot \boldsymbol{d}_{q,m} \boldsymbol{d}_{q,m}^{\mathrm{T}} \right) \boldsymbol{s}_{k,m} \quad (16)$$

represents the reconstructed signal with unit gain. This reduces the subproblem to a linear least-squares problem with a closed-form solution:

$$\hat{\alpha}_q = \frac{\sum_{k,m} \mathrm{tr}\left(\boldsymbol{z}_{q,k,m}^{\mathrm{H}} \boldsymbol{r}_{q,k,m} \right)}{\sum_{k,m} \|\boldsymbol{z}_{q,k,m}\|_{\mathrm{F}}^2}, \quad (17)$$

where $\mathrm{tr}(\cdot)$ denotes the trace operator.

2) Updating Location and Motion Parameters

In contrast to the complex gain, the location parameters $\{r_q, \theta_q\}$ and motion parameters $\{v_q^{\rho}, v_q^{\phi}\}$ are non-linearly embedded in the steering vector $\boldsymbol{a}_q$ and Doppler vector $\boldsymbol{d}_{q,m}$, necessitating numerical optimization. These parameter blocks are updated sequentially within each BCD iteration using gradient descent. At the i-th iteration, the location vector $\boldsymbol{\eta}_q = [r_q, \theta_q]^{\mathrm{T}}$ and motion vector $\boldsymbol{v}_q = [v_q^{\rho}, v_q^{\phi}]^{\mathrm{T}}$ are updated as

$$\begin{aligned} \boldsymbol{\eta}_q^{(i)} &= \boldsymbol{\eta}_q^{(i-1)} - \mu_{\eta} \nabla_{\boldsymbol{\eta}_q} J_q \left(\boldsymbol{\xi}_q^{(i-1)} \right), \\ \boldsymbol{v}_q^{(i)} &= \boldsymbol{v}_q^{(i-1)} - \mu_{v} \nabla_{\boldsymbol{v}_q} J_q \left(\tilde{\boldsymbol{\xi}}_q^{(i)} \right), \end{aligned} \quad (18)$$

where μ_{η} and μ_v are the respective step sizes, and $\tilde{\boldsymbol{\xi}}_q^{(i)} = [\alpha_q^{(i)}, (\boldsymbol{\eta}_q^{(i)})^{\mathrm{T}}, (\boldsymbol{v}_q^{(i-1)})^{\mathrm{T}}]^{\mathrm{T}}$. The gradients with respect to the location vector $\boldsymbol{\eta}_q$ and the motion vector $\boldsymbol{v}_q$ are respectively given by

$$\begin{aligned} \nabla_{\boldsymbol{\eta}_q} J_q &= -2 \sum_{k,m} \Re \left\{ (\boldsymbol{r}_{q,k,m} - \tilde{\boldsymbol{y}}_{k,m})^{\mathrm{H}} \nabla_{\boldsymbol{\eta}_q} \tilde{\boldsymbol{y}}_{k,m} \right\}, \\ \nabla_{\boldsymbol{v}_q} J_q &= -2 \sum_{k,m} \Re \left\{ (\boldsymbol{r}_{q,k,m} - \tilde{\boldsymbol{y}}_{k,m})^{\mathrm{H}} \nabla_{\boldsymbol{v}_q} \tilde{\boldsymbol{y}}_{k,m} \right\}. \end{aligned} \quad (19)$$

The Jacobians $\nabla_{\boldsymbol{\eta}_q} \tilde{\boldsymbol{y}}_{k,m}$ and $\nabla_{\boldsymbol{v}_q} \tilde{\boldsymbol{y}}_{k,m}$ in (19) are constructed from the partial derivatives with respect to each scalar parameter, which can be expressed as

$$\begin{aligned} \frac{\partial \tilde{\boldsymbol{y}}_{k,m}}{\partial r_q} &= \tilde{\boldsymbol{y}}_{k,m} \left(-\jmath 2\pi k \Delta f \frac{2}{c} \right) + \tilde{\alpha}_q \left(\frac{\partial \boldsymbol{H}_{q,m}}{\partial r_q} \right) \boldsymbol{s}_{k,m}, \\ \frac{\partial \tilde{\boldsymbol{y}}_{k,m}}{\partial \theta_q} &= \tilde{\alpha}_q \left(\frac{\partial \boldsymbol{H}_{q,m}}{\partial \theta_q} \right) \boldsymbol{s}_{k,m}, \\ \frac{\partial \tilde{\boldsymbol{y}}_{k,m}}{\partial v_q^{\rho}} &= \tilde{\alpha}_q \left((\boldsymbol{a}_q \boldsymbol{a}_q^{\mathrm{T}}) \odot \frac{\partial (\boldsymbol{d}_{q,m} \boldsymbol{d}_{q,m}^{\mathrm{T}})}{\partial v_q^{\rho}} \right) \boldsymbol{s}_{k,m}, \\ \frac{\partial \tilde{\boldsymbol{y}}_{k,m}}{\partial v_q^{\phi}} &= \tilde{\alpha}_q \left((\boldsymbol{a}_q \boldsymbol{a}_q^{\mathrm{T}}) \odot \frac{\partial (\boldsymbol{d}_{q,m} \boldsymbol{d}_{q,m}^{\mathrm{T}})}{\partial v_q^{\phi}} \right) \boldsymbol{s}_{k,m}. \end{aligned} \quad (20)$$

Here, $\tilde{\alpha}_q = \alpha_q e^{-\jmath 2\pi k \Delta f \frac{2r_q}{c}}$, and $\boldsymbol{H}_{q,m} = \boldsymbol{a}_q \boldsymbol{a}_q^{\mathrm{T}} \odot \boldsymbol{d}_{q,m} \boldsymbol{d}_{q,m}^{\mathrm{T}}$. The detailed expressions for the Jacobians can be obtained through systematic application of the chain rule. We present

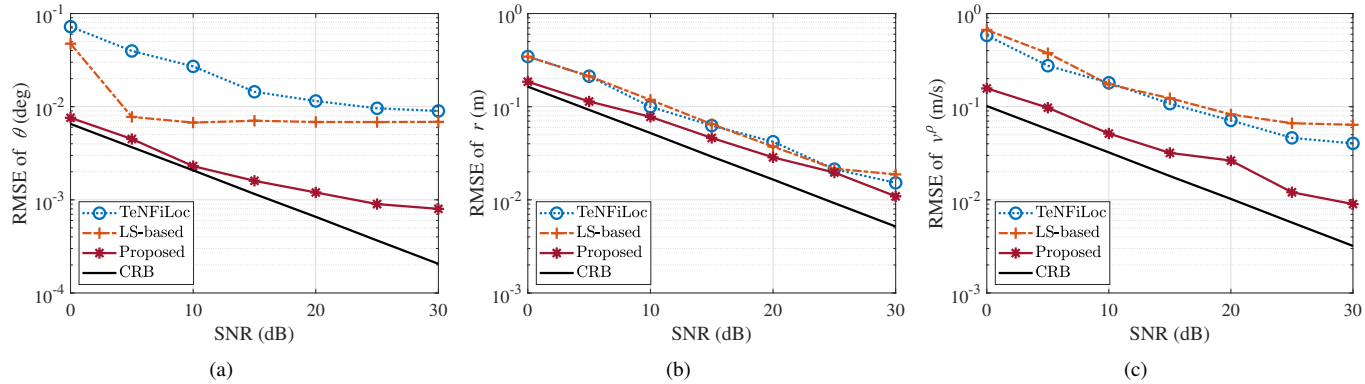

Fig. 2. Comparison of parameter estimation accuracy corresponding to different methods. (a) RMSE of DOA θ. (b) RMSE of range r. (c) RMSE of radial velocity v^{ρ}. Each data point is the result of averaging over $J = 300$ independent Monte-Carlo trials. For reference, the corresponding Cramér–Rao bounds (CRBs) are also plotted for reference.

the derivation for $\partial \boldsymbol{H}_{q,m}/\partial r_q$ as an illustrative example, while the others are omitted due to space constraints.

Specifically, by defining the steering matrix $\boldsymbol{A}_q = \boldsymbol{a}_q \boldsymbol{a}_q^{\mathrm{T}}$ and the Doppler matrix $\boldsymbol{D}_{q,m} = \boldsymbol{d}_{q,m}\boldsymbol{d}_{q,m}^{\mathrm{T}}$, the derivative of $\boldsymbol{H}_{q,m} = \boldsymbol{A}_q \odot \boldsymbol{D}_{q,m}$ with respect to r_q is given by the product rule:

$$\frac{\partial \boldsymbol{H}_{q,m}}{\partial r_q} = \frac{\partial \boldsymbol{A}_q}{\partial r_q} \odot \boldsymbol{D}_{q,m} + \boldsymbol{A}_q \odot \frac{\partial \boldsymbol{D}_{q,m}}{\partial r_q}. \tag{21}$$

The derivatives of the component matrices $\boldsymbol{A}_q$ and $\boldsymbol{D}_{q,m}$ can be further expanded as

$$\frac{\partial \boldsymbol{A}_q}{\partial r_q} = -\jmath \frac{2\pi}{\lambda}\left(\mathrm{diag}\left(\frac{\partial \boldsymbol{\delta}_q}{\partial r_q}\right)\boldsymbol{A}_q + \boldsymbol{A}_q \mathrm{diag}\left(\frac{\partial \boldsymbol{\delta}_q}{\partial r_q}\right)\right), \tag{22}$$

$$\frac{\partial \boldsymbol{D}_{q,m}}{\partial r_q} = \jmath \frac{2\pi m T_{\mathrm{s}}}{\lambda}\left(\mathrm{diag}\left(\frac{\partial \boldsymbol{\nu}_q}{\partial r_q}\right)\boldsymbol{D}_{q,m} + \boldsymbol{D}_{q,m}\mathrm{diag}\left(\frac{\partial \boldsymbol{\nu}_q}{\partial r_q}\right)\right), \tag{23}$$

where $\boldsymbol{\delta}_q = [\delta_{q,1}, \cdots, \delta_{q,N}]^{\mathrm{T}}$ and $\boldsymbol{\nu}_q = [\nu_{q,1}, \cdots, \nu_{q,N}]^{\mathrm{T}}$. The calculation of $\partial \boldsymbol{H}_{q,m}/\partial r_q$ is completed by computing the derivatives of these vectors and substituting the results back into the equations above.

By operating directly on the exact near-field model from (3), the BCD refinement effectively corrects the model mismatch errors from the far-field approximation used in Stage I. As such, the proposed two-stage method balances the computational efficiency and the estimation accuracy, providing a practical solution for near-field localization.

4 Simulation Results

In this section, numerical simulations are presented to validate the effectiveness of the proposed method. The simulation setup involves a scenario with $Q = 2$ UAV targets. The key parameters for the system configuration and target states are summarized in Table I. Based on the system parameters, the boundary between near-field and far-field regions can be calculated as $2D^2/\lambda \approx 86.4$ m, where D is the array aperture. Thus, both UAVs are located in the near-field region.

The parameter estimation accuracy of the proposed method is compared with those of the tensor-based near-field localization (TeNFiLoc) method [23] and the least squares (LS)-based method [14], as shown in Fig. 2. The root mean square error (RMSE) is adopted as the performance metric.

TABLE I: Simulation Parameters

Parameter	Symbol	Value
Carrier frequency	f_c	28 GHz
Subcarrier spacing	Δf	120 kHz
Number of array elements	N	128
Number of targets	Q	2
Ranges of targets	$[r_1, r_2]$	$[11.27, 22.15]$ m
Angles of targets	$[\theta_1, \theta_2]$	$[51.92°, 29.18°]$
Radial velocities	$[v_1^{\rho}, v_2^{\rho}]$	$[22.15, 32.74]$ m/s
Transverse velocities	$[v_1^{\phi}, v_2^{\phi}]$	$[31.98, 37.12]$ m/s

It is clear in Fig. 2 that the proposed method consistently outperforms the competing methods across the entire SNR region for angle θ, range r, and radial velocity v^{ρ}. The reason is that the proposed method is founded on an exact signal model that fully captures the parameter coupling induced by high mobility. In contrast, the competing methods rely on simplified models that omit the Doppler effects of transverse motion. Particularly, the TeNFiLoc method and LS-based method depend on the low-rankness of the signal model to enable decomposition techniques, while the low-rank signal structure can be destroyed by the inclusion of transverse velocity. Since the proposed method is not constrained by this requirement, superior estimation accuracy can be achieved.

5 Conclusion

In this paper, we proposed a two-stage localization method for near-field UAVs. Specifically, coarse parameter estimates are first obtained via a far-field approximation, followed by a high-accuracy refinement using a BCD framework. The utilization of the exact signal model in the refinement stage enables to resolve the complex parameter couplings induced by high mobility of UAV targets. Numerical simulations validate the effectiveness of the proposed method.

6 References

[1] M. L. Rahman, J. A. Zhang, X. Huang, Y. J. Guo, and R. W. Heath, "Framework for a perceptive mobile network using joint communication and radar sensing," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 56, no. 3, pp. 1926–1941, June 2020.

[2] B. Lin, C. Zhao, F. Gao, G. Y. Li, J. Qian, and H. Wang, "Environment reconstruction based on multi-user selection and multi-modal fusion in ISAC," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 15 083–15 095, 2024.

[3] L. Lin, C. Zhou, H. Zheng, Z. Shi, S. A. Vorobyov, and R. W. Heath, "Sensing-aided precoding with high-dynamic moving scatterers," *IEEE Trans. Signal Process.*, pp. 1–16, 2025.

[4] D. An, J. Liu, K. Zhong, J. Hu, H. Yao, H. Li, and F. Gini, "Per-user dynamic controllable waveform design for dual function radar-communication system," *IEEE Trans. Aerosp. Electron. Syst.*, pp. 1–15, 2024.

[5] L. Lin, J. Tong, H. Lin, H. Zheng, and Z. Shi, "Joint target localization and channel estimation for ODDM-ISAC systems," *IEEE Signal Process. Lett.*, vol. 32, pp. 1525–1529, 2025.

[6] Y. Feng, C. Zhao, H. Luo, F. Gao, F. Liu, and S. Jin, "Networked ISAC based UAV tracking and handover towards low-altitude economy," *IEEE Trans. Wireless Commun.*, 2025.

[7] Y. Zhang, H. Shan, H. Chen, L. Cai, Z. Shi, T. Q. S. Quek, and L. Sheng, "Perceptive mobile networks for standalone and cooperative UAV surveillance," *IEEE Trans. Wireless Commun.*, vol. 23, no. 12, pp. 19 916–19 932, Dec. 2024.

[8] M. Wang, X. Li, L. Gao, Z. Sun, G. Cui, and T. S. Yeo, "Signal accumulation method for high-speed maneuvering target detection using airborne coherent MIMO radar," *IEEE Trans. Signal Process.*, vol. 71, pp. 2336–2351, 2023.

[9] X. Shi, C. Yang, W. Xie, C. Liang, Z. Shi, and J. Chen, "Anti-drone system with multiple surveillance technologies: Architecture, implementation, and challenges," *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 68–74, Apr. 2018.

[10] M. Wang, X. Li, D. Chen, M. S. Greco, F. Gini, and T. S. Yeo, "Performance prediction of signal integration for coherent MIMO radar with random amplitude errors," *IEEE Trans. Aerosp. Electron. Syst.*, pp. 1–18, 2025.

[11] C. You, Y. Cai, Y. Liu, M. Di Renzo, T. M. Duman, A. Yener, and A. Lee Swindlehurst, "Next generation advanced transceiver technologies for 6G and beyond," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 3, pp. 582–627, Mar. 2025.

[12] Z. Wang, J. Zhang, W. Yi, H. Xiao, H. Du, D. Niyato, B. Ai, and D. W. K. Ng, "Analytical framework for effective degrees of freedom in near-field XL-MIMO," *IEEE Trans. Wireless Commun.*, vol. 24, no. 4, pp. 3465–3482, Apr. 2025.

[13] H. Lu, Y. Zeng, C. You, Y. Han, J. Zhang, Z. Wang, Z. Dong, S. Jin, C.-X. Wang, T. Jiang, X. You, and R. Zhang, "A tutorial on near-field XL-MIMO communications towards 6G," *IEEE Commun. Surv. Tut.*, vol. 26, no. 4, pp. 2213–2257, 2024.

[14] J. Fang, H. Chen, W. Liu, S. Yang, C. Yuen, and H. C. So, "Three-dimensional localization of mixed near-field and far-field sources based on a unified exact propagation model," *IEEE Trans. Signal Process.*, vol. 73, pp. 245–258, 2025.

[15] B. Zhao, C. Ouyang, Y. Liu, X. Zhang, and H. V. Poor, "Modeling and analysis of near-field ISAC," *IEEE J. Sel. Topics Signal Process.*, vol. 18, no. 4, pp. 678–693, May 2024.

[16] M. Wang, X. Li, L. Gao, X. Li, Z. Sun, G. Cui, and T. S. Yeo, "Multichannel signal integration approach for high-speed target detection in coherent MIMO radar system," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 6, pp. 8298–8315, Dec. 2024.

[17] C. Giovannetti, N. Decarli, and D. Dardari, "Performance bounds for velocity estimation with extremely large aperture arrays," *IEEE Wireless Commun. Lett.*, vol. 13, no. 12, pp. 3513–3517, Dec. 2024.

[18] X. Zhang, H. Zhang, and Y. C. Eldar, "Near-field sparse channel representation and estimation in 6G wireless communications," *IEEE Trans. Commun.*, vol. 72, no. 1, pp. 450–464, Jan. 2024.

[19] Z. Zhu, R. Yang, C. Li, Y. Huang, and L. Yang, "Adaptive joint sparse Bayesian approaches for near-field channel estimation," *IEEE Trans. Wireless Commun.*, vol. 24, no. 3, pp. 2590–2605, Mar. 2025.

[20] W. Zuo, J. Xin, N. Zheng, and A. Sano, "Subspace-based localization of far-field and near-field signals without eigendecomposition," *IEEE Trans. Signal Process.*, vol. 66, no. 17, pp. 4461–4476, Sep. 2018.

[21] J. He, T. Shu, L. Li, and T.-K. Truong, "Mixed near-field and far-field localization and array calibration with partly calibrated arrays," *IEEE Trans. Signal Process.*, vol. 70, pp. 2105–2118, 2022.

[22] X. Wu, W.-P. Zhu, and J. Yan, "Atomic norm based localization of far-field and near-field signals with generalized symmetric arrays," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, Barcelona, Spain, May 2020, pp. 4762–4766.

[23] I. Podkurkov, G. Seidl, L. Khamidullina, A. Nadeev, and M. Haardt, "Tensor-based near-field localization using massive antenna arrays," *IEEE Trans. Signal Process.*, vol. 69, pp. 5830–5845, 2021.

[24] H. Miao and M. Peng, "Near-field high-speed user sensing in wideband mmWave communications: Algorithms and bounds," *IEEE Trans. Signal Process.*, vol. 73, pp. 919–935, 2025.